Mesopore size impact on TENG

"How does mesopore size affect TENG charge output" *Generated by LeapSpace*

Mesopore size affects TENG charge output in a non-linear way: smaller-to-moderate mesopores generally improve charge output, while overly large pores reduce it because they lower surface area and charge-trapping efficiency. In the retrieved evidence, the best-performing structures are consistently **well-developed mesoporous networks with relatively small pores**, not large voids [46] [92] [100] [101] [102] [103] [104].

What the evidence shows

Mesopore size / structure	Effect on TENG charge output	Why it matters
Small mesopores / nanoporous mesostructures	**Higher charge output**	They provide more internal surface, more charge-trapping sites, and stronger electrostatic induction [46] [101] [103]
Mesopores in the low-nanometer range	**Strong output enhancement**	Mesoporous structures around **3.6 and 4.9 nm** were associated with high surface area and improved charge transfer in a TENG-active material [100]
~20–40 nm pores in porous SiO_2	**High triboelectric output**	Rice-husk SiO_2 with **20–40 nm** pores produced much higher output than flat SiO_2, with a strong current density and high area power density [92]
Larger pores / overly open structures	**Lower output**	Large pores reduce surface area-to-volume ratio and trap fewer charges, which weakens triboelectric performance [102] [104]

Practical takeaway

For **TENG charge output**, the most supportable target is **controlled mesoporosity in the small-to-medium range**, not the largest possible mesopores [92] [100] [102]. The mechanism is consistent across studies: **smaller pores increase contact area and charge-trapping density**, while **larger pores start to hurt output because the effective electrified area drops** [46] [101] [104].

Bottom line

If you want higher TENG charge output, aim for well-ordered mesopores that are small enough to maximize surface area and trapping, but not so small or dense that the structure loses accessibility. The retrieved evidence points to **small mesopores / low-tens-of-nanometers pores as the most effective regime**, with performance falling off as pores become too large [92] [100] [102].

Confidence level: **High** — multiple independent TENG studies consistently link smaller, well-controlled mesoporosity with higher charge output, and several sources explicitly show reduced performance when pores become larger [46] [92] [100] [101] [102] [104].

References

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